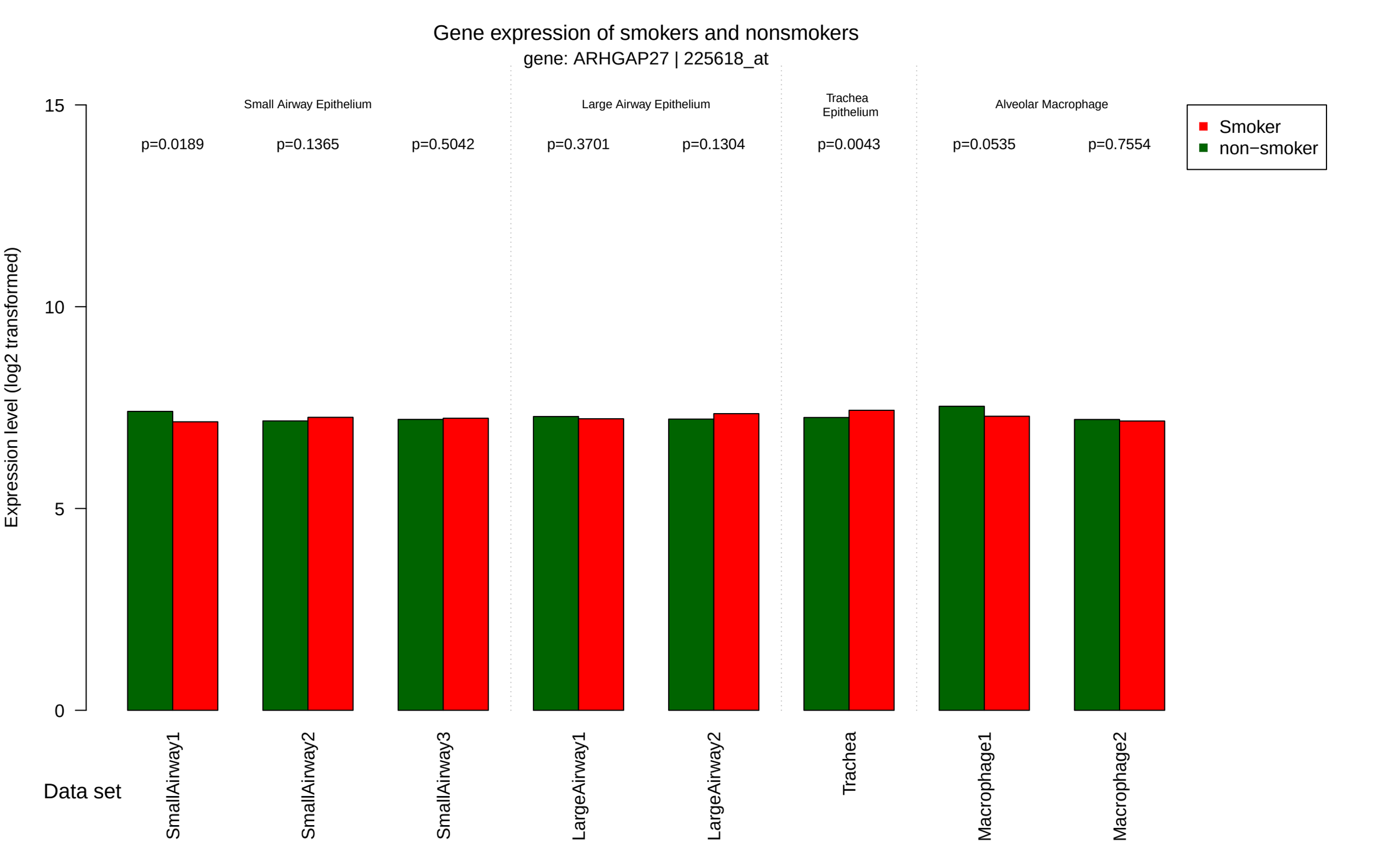
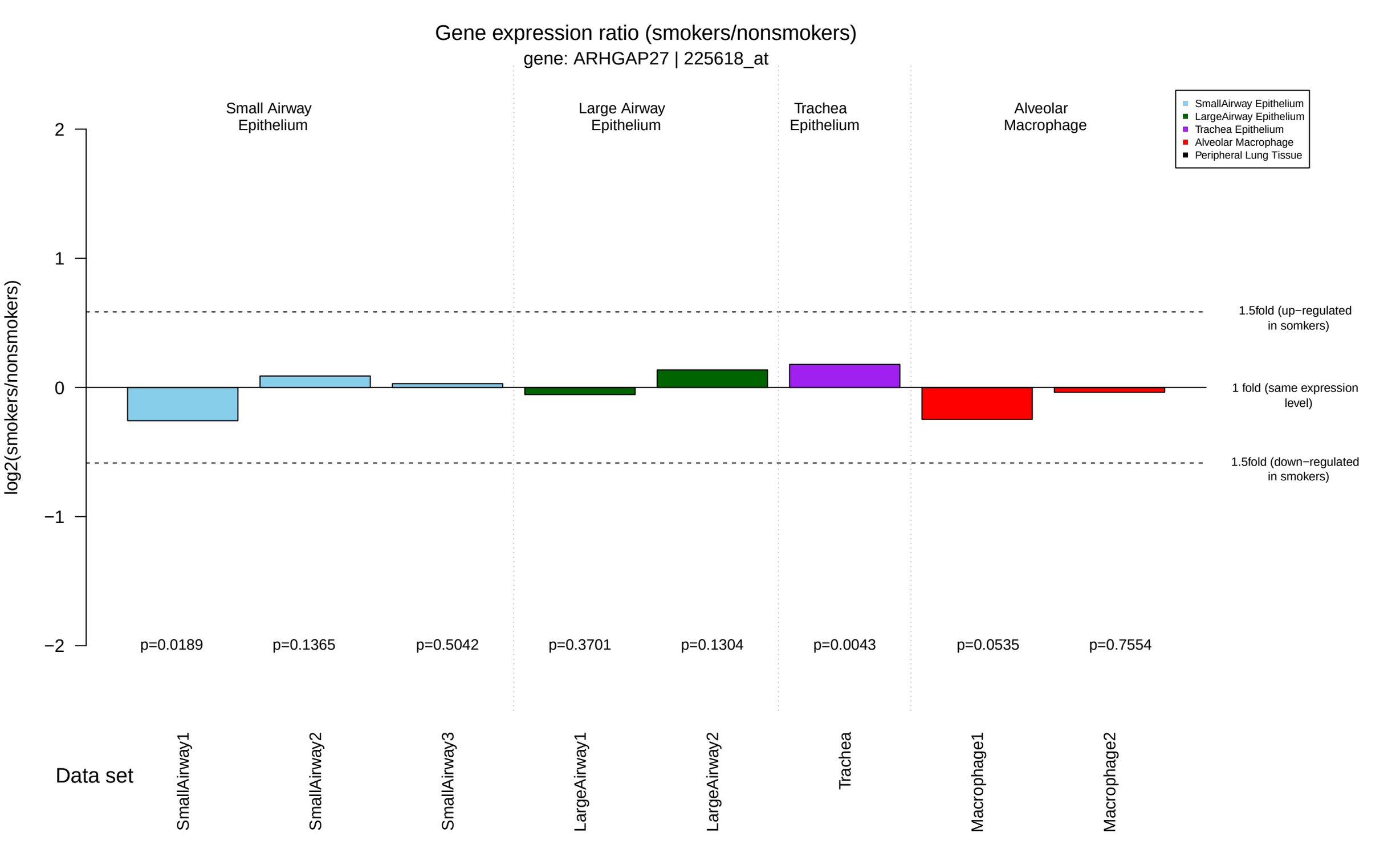
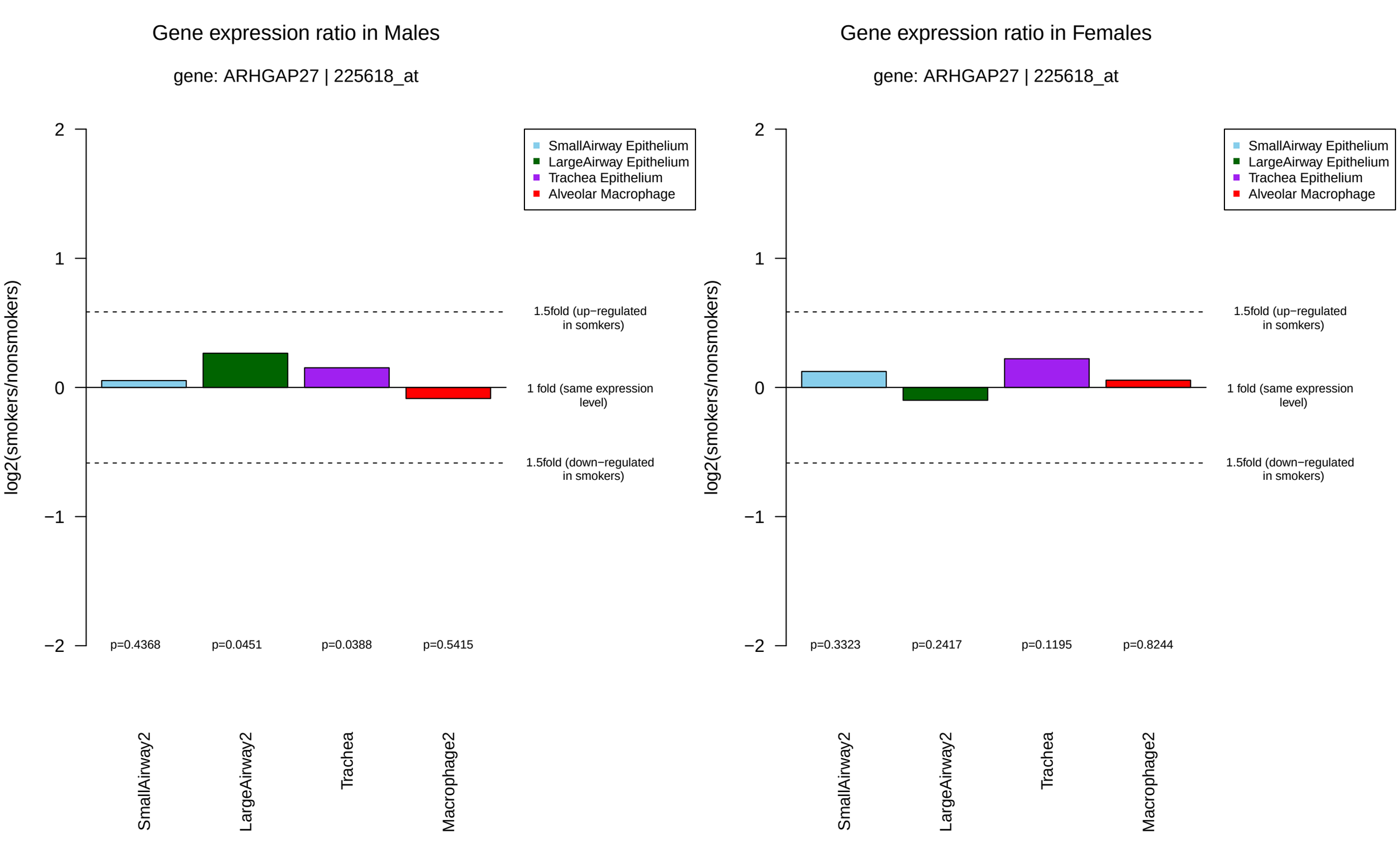
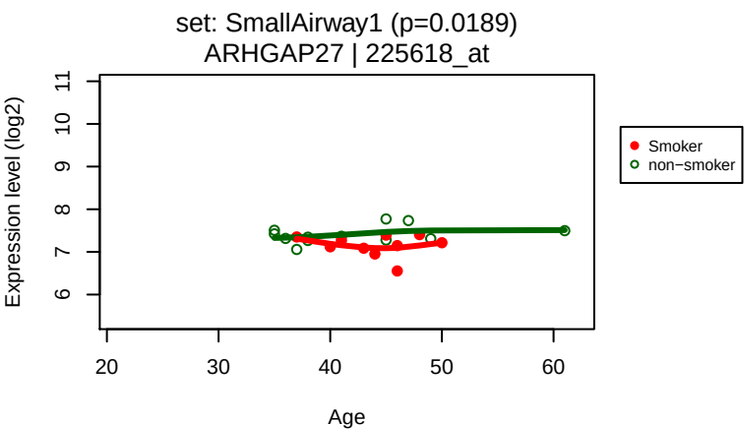
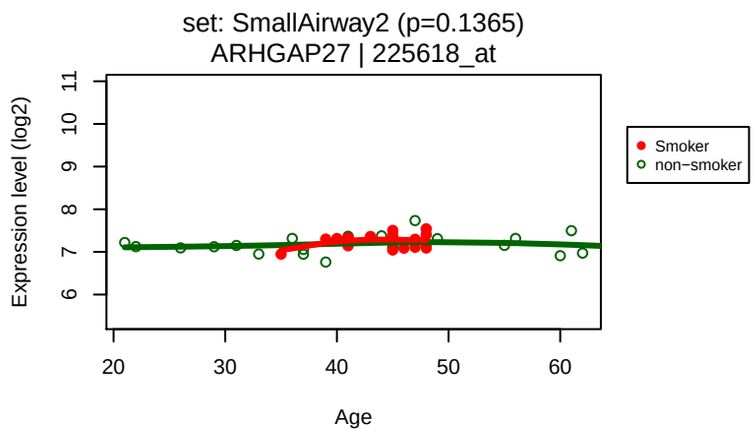
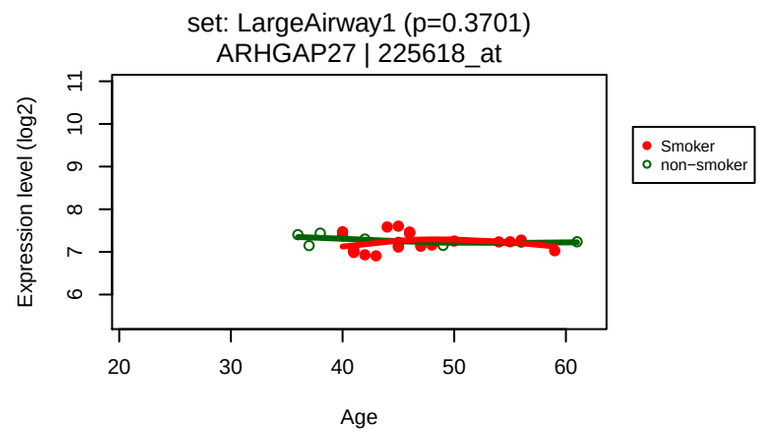

Gene expression of smokers and nonsmokers
gene: ARHGAP27 | 225618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0189
p=0.1365
p=0.5042
p=0.3701
p=0.1304
p=0.0043
p=0.0535
p=0.7554
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

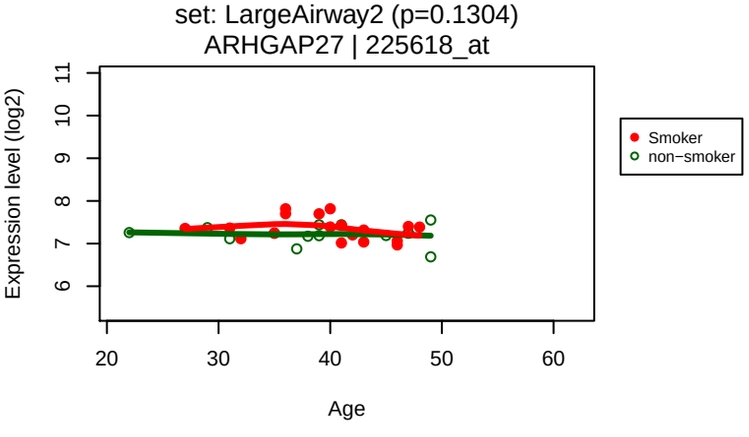

Gene expression ratio (smokers/nonsmokers)
gene: ARHGAP27 | 225618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0189
p=0.1365
p=0.5042
p=0.3701
p=0.1304
p=0.0043
p=0.0535
p=0.7554
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

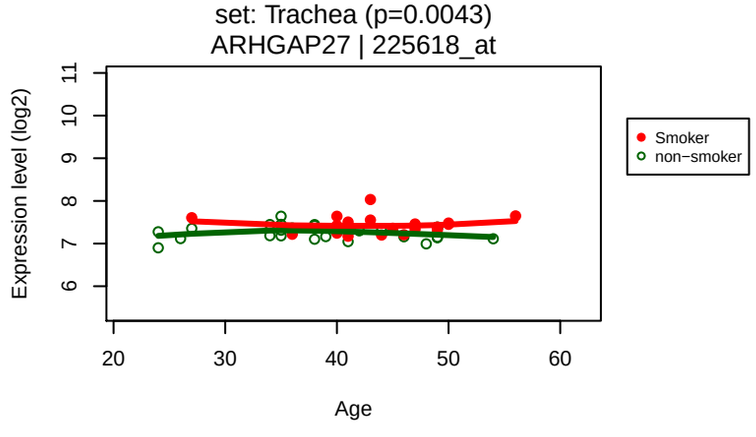

Gene expression ratio in Males
gene: ARHGAP27 | 225618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4368
p=0.0451
p=0.0388
p=0.5415
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARHGAP27 | 225618_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3323
p=0.2417
p=0.1195
p=0.8244
SmallAirway2
LargeAirway2
Trachea
Macrophage2

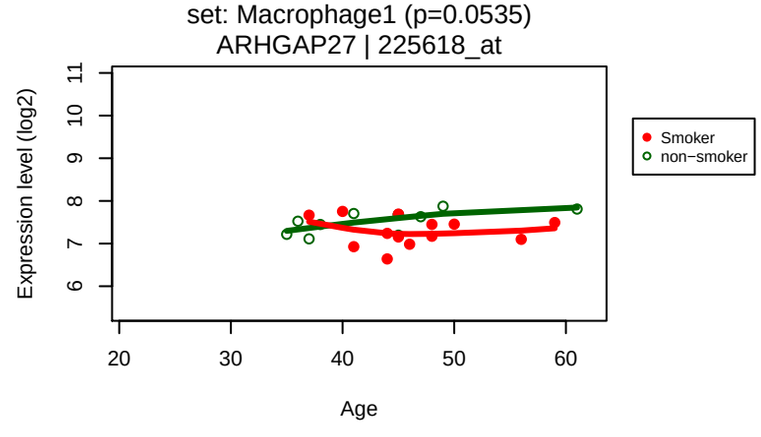

set: SmallAirway1 (p=0.0189)
ARHGAP27 | 225618_at
Expression level (log2)
Age
Smoker
non−smoker

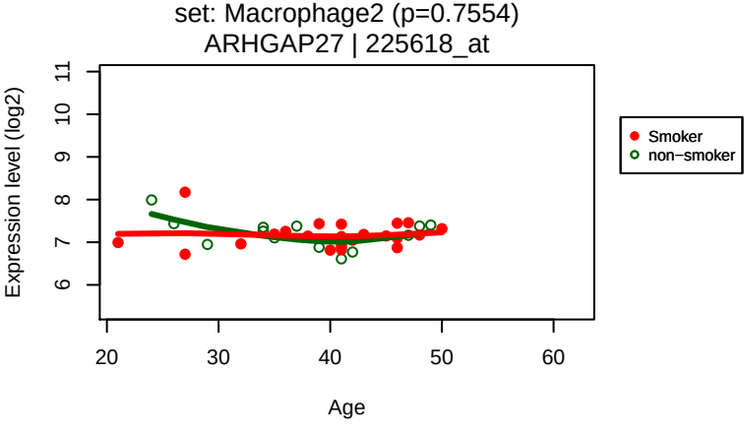

set: SmallAirway2 (p=0.1365)
ARHGAP27 | 225618_at
Expression level (log2)
Age
Smoker
non−smoker

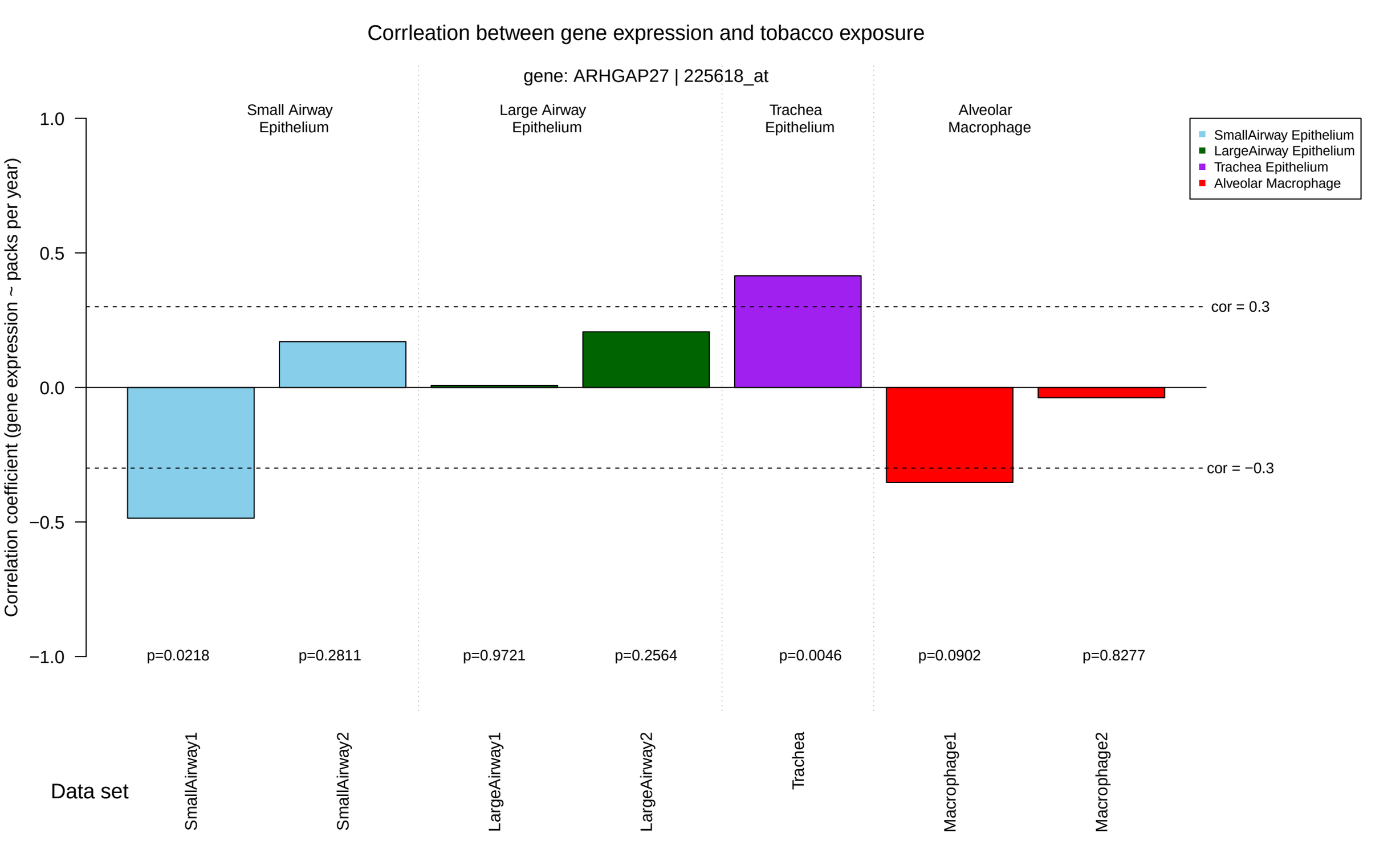

set: LargeAirway1 (p=0.3701)
ARHGAP27 | 225618_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1304)
ARHGAP27 | 225618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0043)
ARHGAP27 | 225618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0535)
ARHGAP27 | 225618_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7554)
ARHGAP27 | 225618_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARHGAP27 | 225618_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0218
p=0.2811
p=0.9721
p=0.2564
p=0.0046
p=0.0902
p=0.8277
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2